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Development of a new method for maintainability and downtime analysis of mining machinery

M. Rezaei Dashtaki, A. Jandaghi Jafari & S. H. Hoseinie✉

Shovels are universally acknowledged as vital machinery in surface mining production systems, and enhancements to this equipment substantially augment operational mine output capacity. Extending the length of continuous operation is a cost-efficient strategy that may result in significant productivity improvements, as it is directly linked to the fleet's reliability and maintainability. This research analyses the maintainability of a fleet of ten rope shovels in an open-pit copper mine, consisting of three separate models. To this end, repair data for all ten shovels over two years were analyzed, and maintainability models were assigned to each shovel. The results indicate that all shovels are predominantly maintained in less than 45 h, and 80% of repairs can be accomplished within a single shift, which is operationally understandable and manageable for the entire crew. Alongside maintainability research, a novel quantitative method has been established to examine the composition of downtimes based on primary categories of action: pre-repair actions, repair actions, and post-repair actions. Analysis of 25 mechanical and 25 electrical failures revealed that repair duration was almost 50% of the overall downtime. The residual time was allocated to multiple tasks, including vehicle arrival, delays, requisite preparations, diagnostics, and performance testing and validation. Approximately 30% of the overall downtime stemmed from the duration needed for transportation from the repair facility to the machine and operational delays. The findings established a foundation for recommending a viable approach to enhance the maintenance and repair conditions of the Shovel fleet at the copper mine under consideration.

Keywords Shovel, Repair action, Availability, Maintenance

Nowadays, in the complex global minerals industries market, due to high competition and dynamic supply-demand trends, it is crucial for the companies' directors to strategically plan and efficiently implement their resources through lean processes to get higher business values. Even though the mineral reserve is the most valuable resource of a mining company, a large fleet of capital-intensive machinery is known as the most valuable asset of a mining company. Thus, maintenance plays a crucial role across the mines as it directly impacts the machinery performance, fleet operational capacity, and mine productivity^{1–3}.

As the most important machine in the mine operation chain, shovels directly affect the mine production rate and machinery fleet productivity. Therefore, boosting the shovel availability in a cost-effective way under an optimized maintenance program could assure its performance along with the whole ore delivery to the mine crusher.

Availability in heavy machinery can be attributed to two key factors: reliability and maintainability. Reliability refers to various machine components' design and material choices that lead to equipment durability. At the same time, maintainability is associated with repairing and maintaining infrastructure, involving numerous factors such as personnel capabilities, technology, tools, and more^{4–6}.

One effective method for optimizing availability is analyzing repair processes and identifying bottlenecks in maintenance and repair procedures to minimize maintenance lead time^{7,8}. In this article, a case study was conducted on several shovels in a large copper mine (Sarcheshmeh Copper Mine in Iran) to analyze the primary components of maintenance time based on a series of unit activities, to present the findings for potential improvements in the mine's overall availability and efficiency. The primary objective of this paper is to assess the maintainability of the Shovel fleet in the Sarcheshmeh copper mine and provide recommendations for its enhancement. The findings of this research will contribute to the development of more efficient maintenance

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strategies for the Sarcheshmeh copper mine's shovel fleet, potentially increasing production capacity and overall mine efficiency.

To optimize equipment availability, previous studies have focused on maintainability analysis and downtime reduction, generally classifying downtime into four primary components: Mean Detection Time, Mean Decision-making time, Mean time to repair, and Mean Function Test time. Our research expands upon this conventional downtime classification by introducing a more detailed analysis that includes the following additional components: Time to reach the failed machine, Failure detection time, Time to decide about a repair action, Preparation time to start repair operation, Time to perform the main repair operation, and Functional testing of repair performance and Delays.

By conducting a comprehensive case study on shovels in the Sarcheshmeh Copper Mine in Iran, we demonstrate the value of this enhanced downtime classification in achieving more accurate analyses, better allocation of human resources, and improved maintainability. Our study contributes to the existing body of knowledge by offering a detailed approach that addresses the limitations of previous maintainability analyses and has practical implications for mining companies seeking to optimize their operations.

Maintainability and downtime analysis; definitions and review

Maintainability is a key aspect of system design and performance, referring to the ability of a system to undergo necessary repairs and maintenance to sustain optimal functionality. This concept encompasses not only the time required for repairs but also the quality and ease of maintenance processes. In today's rapidly advancing technological landscape, a focus on maintainability can significantly impact cost reduction and extend the useful life of systems.

In this section, we will examine past studies related to downtime analysis. This analysis helps us identify weaknesses in current processes and propose improvements. Given the critical importance of downtime in overall system performance, accurately identifying and analyzing the factors influencing it can lead to enhanced maintainability and improved system efficiency. These efforts not only reduce operational costs but also increase user satisfaction and overall organizational productivity.

Maintainability

Maintainability refers to the capacity of an item to be preserved or returned to a specific condition when maintenance is performed. Maintainability is an inherent characteristic of engineering machinery that focuses on the ease, cost-efficiency, and safety of maintenance procedures. As most engineered products require maintenance throughout their life cycle, it is crucial to design them in a way that facilitates safe and efficient maintenance. This involves minimizing the time, resources (e.g., labor, materials, equipment, and facilities), and environmental impact associated with maintenance activities. A well-designed product with high maintainability contributes to overall system reliability, cost savings, and reduced downtime during its operational lifespan^{9,10}.

Numerous elements significantly affect the maintainability index, such as requisite tools, essential materials, personnel capabilities at maintenance facilities, and a myriad of other contributing factors. A notable development occurred in 1997 when Ebeling¹¹ presented an elaborate classification of factors influencing maintainability, which has since served as a valuable resource for engineers and researchers.

Figure 1 illustrates maintainability taxonomy, providing a methodical framework for understanding and evaluating the diverse range of elements that impact maintainability in various engineering products or systems. This enables professionals to systematically identify areas for improvement and devise strategies that optimize maintainability. Consequently, such a proactive approach fosters more efficient maintenance practices, minimizes downtime, and ultimately enhances overall system reliability^{11,12}.

Ben Daya et al.⁸ identified key elements significantly impacting maintainability, namely: Accessibility, Simplification, Standardization and interchangeability, Modularization, Identification and labeling, Testability and Diagnostic Techniques, Human Factors, and Environmental Factors. These factors emphasize the importance of adopting a multifaceted approach to improve maintainability in various systems and industries. Enhancing accessibility, simplifying designs, standardizing components, promoting modularization, and ensuring proper identification and labeling all contribute to streamlining maintenance processes. Additionally, focusing on human factors and the work environment fosters an efficient maintenance culture.

The maintainability function is defined by Eq. (1) as follows⁷:

$$m(t) = \int_0^t f_r(x) dx \quad (1)$$

Where t represents the time, $f_r(t)$ represents the probability density function of the repair time, and $m(t)$ represents the maintainability function.

Maintainability is a key factor in boosting the mining machinery availability because mines are mostly far from industrial centers and maintenance support and logistics affect the maintainability and fleet availability significantly. Prior researchers have investigated and modeled the maintenance capabilities of various drilling machines, such as underground drilling machines¹², rotary drilling machines⁹, Tunnel Boring Machines (TBM)¹³, and shearer loaders¹⁴. The availability of standardized criteria, particularly DOD-HNDBK-791¹⁵, has significantly contributed to the accurate calculation and scientific application of this essential parameter for large-scale machinery.

Specifically in the mining machinery context, in 2011, Hoseinie et al.¹⁴ conducted a study on the reliability and maintainability of the electrical system of a drum shearer. Analyzing 19 months of maintenance and failure data, they determined the statistical distribution of failures and calculated the optimal preventive maintenance



Fig. 1. Affecting factors on maintainability and their internal interactions.

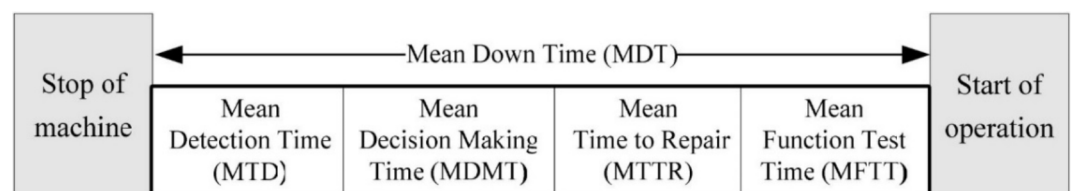


Fig. 2. The different states associated with failure occurrence¹⁶.

intervals for varying system reliability levels. The study also revealed a rapid repair time for the shearer's electrical system, with 80% of failures resolved within 1.45 h.

In 2013, Rahimdel et al.⁹ explored the reliability and maintainability of the pneumatic system in rotary drilling machines to enhance drilling efficiency and safety. Data from four drilling machines identified optimal times for preventive maintenance and average repair times for pneumatic system failures in each machine. These findings offer valuable insights for optimizing maintenance strategies in rotary drilling machines.

Roy et al.¹⁰ examined the maintainability of electric rope shovels in a mining operation to improve productivity. Failure and repair data from four 10 m³ shovels were analyzed, and dipper and electrical subsystems were identified as weak points due to higher failure rates. Reliability dropped below 50% after 24 h for all shovels. The authors recommended preventive maintenance schedules and dynamic inspection frequencies to reduce downtime and increase productivity.

Downtime analysis approaches

Downtime analysis is vital for optimizing operational efficiency across industries. Key approaches include Pareto Analysis, which identifies significant causes of downtime for prioritization; Root Cause Analysis (RCA) to pinpoint underlying factors and develop corrective actions; Failure Modes and Effects Analysis (FMEA) as a proactive approach for anticipating potential failure modes; Overall Equipment Effectiveness (OEE) to evaluate performance based on availability, performance, and quality; Reliability-Centered Maintenance (RCM) for tailored strategies to optimize equipment reliability; and Data Analytics with Predictive Maintenance utilizing historical data to monitor equipment health, predict failures, and proactively schedule maintenance activities.

Numerous scholarly studies have investigated failure times, repair times, Maintainability, and associated elements across various industries. A notable example is the research conducted by Ben-Daya et al.⁸ assessed and defined key components such as failure times, repair times, and the time from failure to repair. In a related study, Barabady¹⁶ defined four critical aspects of the failure-to-repair process: Mean Detection Time (MDT), Mean Decision Making Time (MDMT), Mean Time to Repair (MTTR), and Mean Function Test Time (MFTT). In Fig. 2, the defined elements for the time from failure to repair. These components span the stages

of failure detection, decision-making, repair, and functionality testing within the maintenance process. MDT represents the time required to detect a failure, which can be minimized by improving detection methods and implementing real-time monitoring. MDMT signifies the time taken for decision-making, which can be reduced through efficient communication and decision-making processes. MTTR focuses on the time taken for repairs, including diagnosis, disassembly, parts replacement, reassembly, and testing. This time can be shortened via technician training, spare parts management, and predictive maintenance techniques. MDMT refers to the average time required for functionality testing after a repair, which can be optimized through improved test procedures, automated testing tools, and investment in quality assurance practices.

In this paper, not only the mathematical approach is applied to model the maintainability of cable shovels but also a new hybrid (qualitative and quantitative) approach is developed and applied for downtime analysis. The details of this approach will be presented in the coming parts of the paper.

Shovel fleet maintainability in Sarcheshmeh copper mine

Sarchemeh Copper Mine, the largest open pit mine in Iran and one of the world's top 15 largest copper mines, produces 31 million tones of ore annually and is planning to produce 68 million tonnes in the coming five years of development. The copper sulfide ore is mainly by 10 rope shovels with 12 and 15 m³ bucket capacity. To increase fleet availability and enhance operation productivity, a comprehensive study was carried out on mine shovels which generally followed a hybrid quantitative and qualitative analysis.

In this section, we have measured the maintainability of the shovels used in the Sarcheshmeh copper mine, utilizing the insights and classifications introduced in Sect. 2. A new classification framework for downtime analysis has been proposed, allowing for a comprehensive evaluation of the downtime experienced by these shovels. By systematically analyzing the factors contributing to downtime, we aim to identify specific areas for improvement, ultimately enhancing the operational efficiency and reliability of the shovels in this critical mining environment. This approach not only contributes to better maintenance practices but also supports the overall productivity of mining operations at Sarcheshmeh.

Maintainability analysis

Maintainability studies necessitate a wide range of data which was acquired from the database of failure recording and repair management system utilized at the mine's central repair shop. To collect this data, the database was queried to provide all shovel failures from 1st March 2022 to 1st March 2023. Each shovel's documented failures, repair actions, applied spare parts, and total downtime were classified and adjusted accordingly. Due to the high level of importance of shovels in the mine production chain, the maintenance operation usually costs and is manpower-extensive and many actions are carried out parallel (Fig. 3).

After refining and analyzing the primary data, the information was inputted into the Easyfit software for statistical analysis. The Anderson-Darling goodness-of-fit index was utilized to assess the failure data for each shovel and identify the statistical behavior using distribution functions such as Weibull (two- and three-parameter), normal, lognormal, exponential, and logistic. The results of the data analysis and selected model are presented in Table 1. Following the statistical analysis, maintainability curves were plotted for each shovel, as presented in Figs. 4 and 5.

The data indicated that SH10 and SH08 shovels demonstrated the highest maintainability among all shovels in the mine. 80% of the repairs of these two shovels were completed in less than 130 min (two hours and ten minutes), which is considerably efficient for such huge machinery. Conversely, shovels SH03 and SH04 show the



Fig. 3. Shovel repair operation in Sarcheshmeh Copper Mine.

Shovel code	Number of recorded failures	Average downtime (h)	Best-fitted statistical distribution	Probability density function parameters
SH01	146	10.9	Lognormal	$\mu = 0.1063, \sigma = 1.5$
SH02	310	4.1	Lognormal	$\mu = 0.1467, \sigma = 1.342$
SH03	203	18.9	Lognormal	$\mu = 0.4145, \sigma = 1.612$
SH04	171	2.7	Weibull 2P	$\alpha = 0.8298, \beta = 1.484$
SH05	390	4.3	Lognormal	$\mu = 0.0232, \sigma = 1.22$
SH06	344	5.4	Lognormal	$\mu = 0.0183, \sigma = 1.356$
SH07	146	5.9	Weibull 2P	$\alpha = 0.6615, \beta = 2.373$
SH08	358	2	Log-normal	$\mu = 0.161, \sigma = 1.101$
SH09	299	2.8	Log-normal	$\mu = 0.0022, \sigma = 1.243$
SH10	209	4.5	Weibull 2P	$\alpha = 0.8056, \beta = 1.243$

Table 1. Results of statistical analysis of the collected downtime data of each shovel.

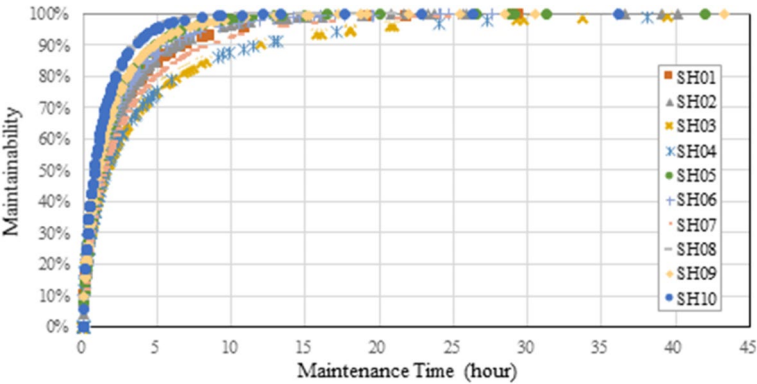


Fig. 4. Maintainability curve for the shovel fleet at Sarcheshmeh copper mine (all failures).

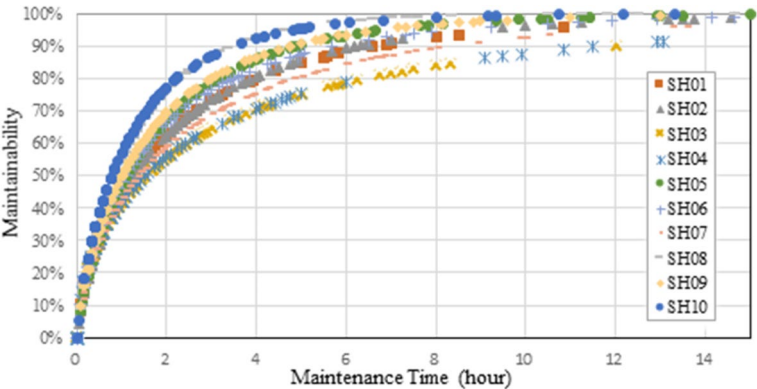


Fig. 5. Maintainability curve for the shovel fleet at Sarcheshmeh copper mine (repairs within less than 15 h).

lowest maintainability, with 80% of their breakdowns lasting approximately 180 min. The maintainability of the other shovels in the mine varied between these four extreme plots.

As shown in Fig. 4, all the recorded repairs for the shovel fleet in Sarcheshmeh Copper Mine were completed within 45 h, with only 20% of repairs in the worst-case scenario exceeding 7 h. It is important to note that the average repair time presented in Table 1 is derived from data collected over a year at the mine, considered as the statistical population. The modeling conducted represents the average repair time for this specific data set. As previously stated, all failures were repaired in under 45 h, with 80% completed within 7 h. For further details, Fig. 5 illustrates the maintainability of failures repaired in less than 15 h.

According to Figs. 4 and 5 obtained from statistical modeling, it can be inferred that considering the mine's operational shifts and the productive working hours of each shift, 80% of all repairs to the mine shovels can be completed within a single shift which is operationally understandable and planable for all crew simply. The remaining 20% of repairs require between one to seven shifts to be addressed. This information can be used

to optimize maintenance schedules, workforce allocation, and spare parts inventory management, ultimately enhancing the overall efficiency of the mining operation.

Downtime analysis

The second phase of the field study was focused on the analysis of the detailed operational tasks and elements that are carried out to conduct repairs and send the machine back to operation. These parameters are primary factors contributing to downtime, along with assessing the maintainability. For this purpose, a new multi-attribute timing regime is developed considering the field maintenance challenges in large open pit mines to facilitate the downtime study in detail as shown in Fig. 6. The aim is to obtain more detailed insights into the primary factors influencing the time from failure to repair and the return of the equipment to the production cycle by the repair workshop.

As Fig. 6 reveals, there are some pre-repair actions, main repair operations, and some post-repair actions along with unexpected delays that could occur in any part of the downtime duration. The desired factors mainly consider the workshop's support and logistics, personnel skills, tools, platform performability, 5 S, environmental factors, and managerial aspects.

It is important to notice that the presented methodology tries to help the operation and maintenance engineers understand the downtime better and optimize the resource allocation as well. Achieving these two important outcomes of this method leads to the maintenance operation of mining machinery to lower costs, especially opportunity costs caused by production loss during downtime.

To study the shovel downtimes in Sarcheshmeh mine deeply and significantly, it was decided to separate the failures into mechanical and electrical types. This strategy was also developed because of the existence of some differences between these two types of maintenance in crew, tools, and support.

During the study, fifty shovel failures and related downtimes, including 25 electrical and 25 mechanical issues, were precisely monitored and all seven mentioned time shares in Fig. 6 were measured and recorded by stopwatch and recorded over 40 days. The timing of the downtimes was initiated once the shovel operator reported the failure to the workshop emergency center until the equipment was fully functional and back in operation again. Figures 7 and 8, and 9 present the result of the downtime analysis of mechanical, electrical, and all (both mechanical and electrical) failures respectively. This analysis helps identify potential areas for improvement, such as optimizing response times, streamlining repair procedures, and ensuring the availability of necessary spare parts. Addressing these factors can help minimize downtime and enhance the overall maintainability of the shovel fleet. Showcase the findings from the field investigations of the repair work performed during breakdown occurrences.

Figure 7 reveals that only 54% of the mechanical failures' downtime at the Sarcheshmeh copper mine, is attributed to a direct repair operation and fixing the malfunction. The remaining 46% is dedicated to dealing with subordinate issues and maintenance tasks. Among the seven components evaluated, the arrival time, preparation time, and delays are classified as support parameters, while diagnosis time, decision time, repair time, and repair verification time are more closely linked to the competence of employees and their experience.

Notably, 37% of the repair time was spent on support tasks, while 63% was allocated to other productive tasks. The actual time dedicated to diagnosis and decision-making (4% of the total time) is notably low compared to the other assessed aspects. This finding highlights the exceptional expertise and extensive knowledge of the mechanical maintenance crew working on the shovels in the Sarcheshmeh mine. Furthermore, the distant location of the shovels indicates the necessity for additional resources, such as light vehicles, drivers, and semi-heavy to heavy vehicles, to facilitate operations and provide assistance during repair and maintenance operations. By addressing these factors and optimizing resource allocation, mine operators can potentially reduce downtime and enhance overall productivity.

The chart demonstrates a decrease in the direct time spent on repairs to 44%, alongside a 5% shift in the proportion of activities related to electrical repairs. This change can be attributed to the relative ease of performing mechanical repairs compared to electrical maintenance work. The 54% share for mechanical maintenance is considered an acceptable difference.

Figure 8 which reveals the downtime composition in electrical failures of shovel fleet, realizes some new facts about the case studied operation. A notable finding is that electrical issues require significantly more time for diagnosis and identification (10%), representing an increase of over 50% in comparison with mechanical failures.

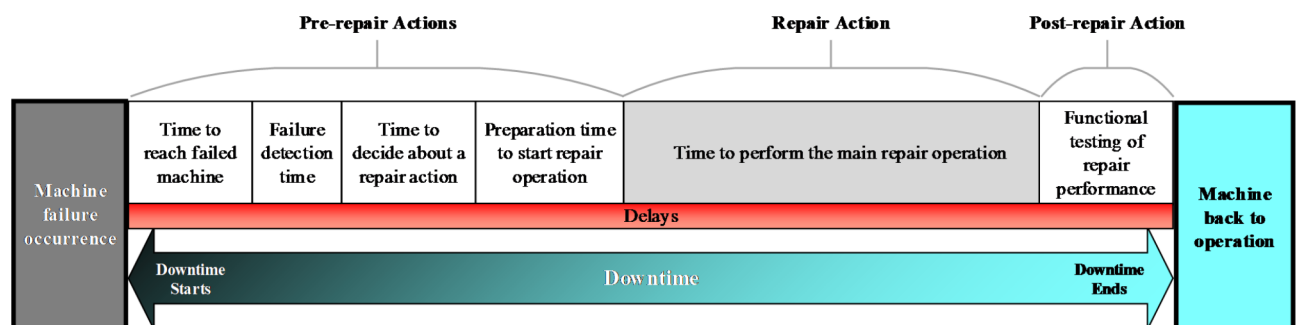


Fig. 6. Composition of downtime of mining machinery applied in this research.

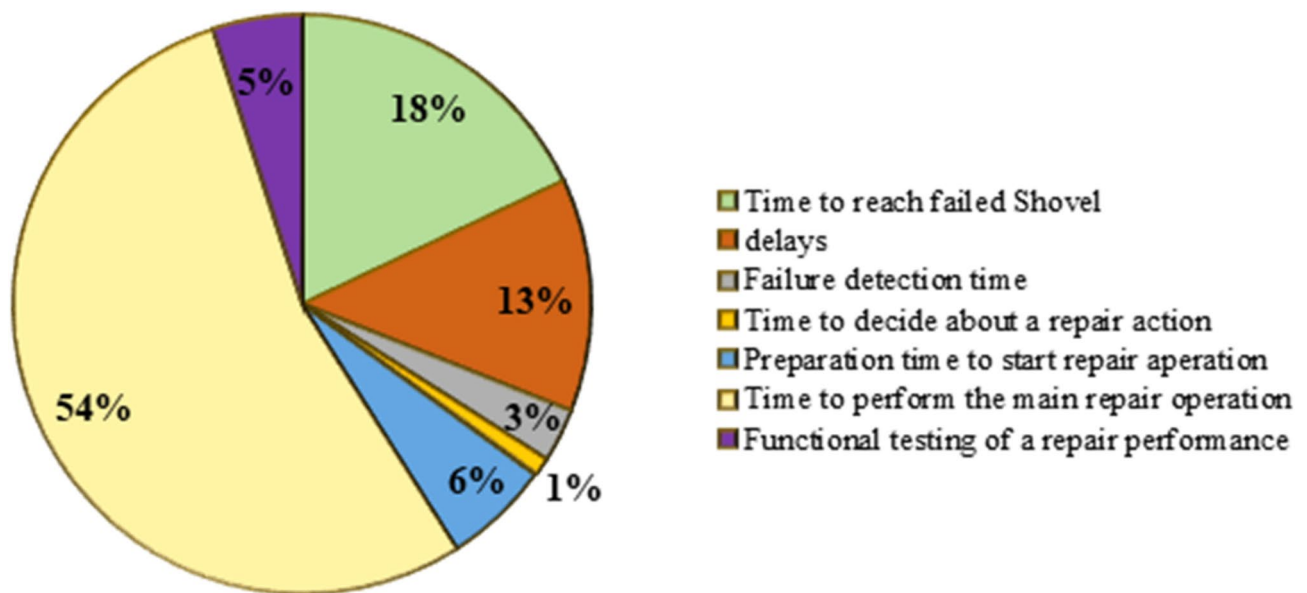


Fig. 7. Composition of mechanical failures' downtimes of the shovel fleet at Sarcheshmeh copper mine.

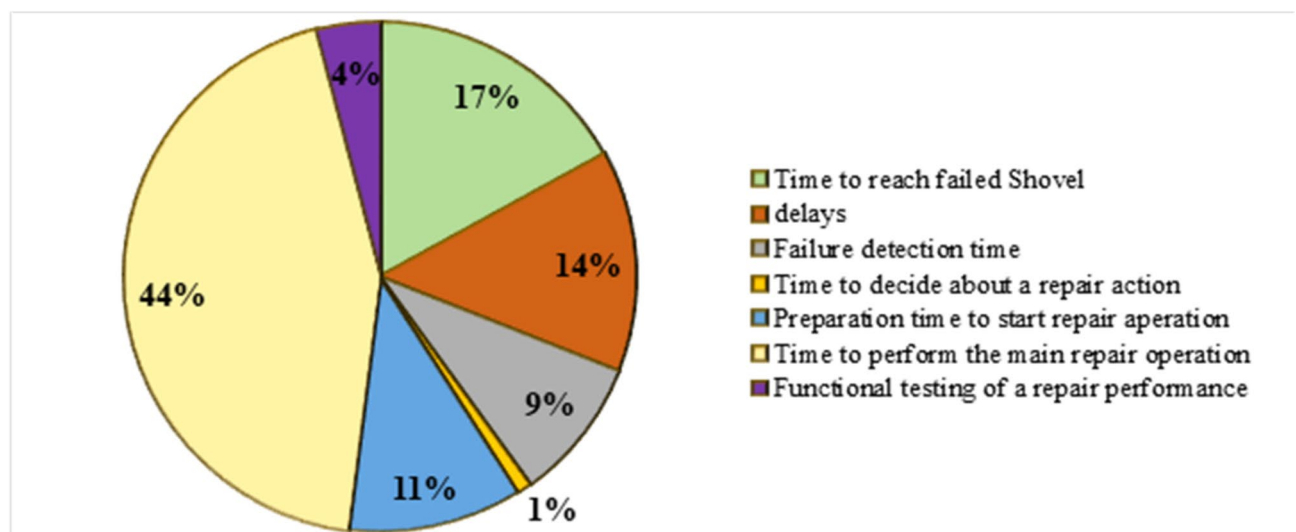


Fig. 8. Composition of electrical failures' downtimes of the shovel fleet at Sarcheshmeh copper mine.

This discrepancy arises because electrical defects often do not produce subtle audio or visual signals that must be detected by the operator or service personnel. This makes accurate problem identification more challenging and time-consuming. After accounting for the time allocated to direct repairs, it is observed that 42% of the machine downtime for electrical maintenance is attributed to factors related to support and logistic delays.

By combining the whole data from 50 downtime analysis cases as a black box (Fig. 9), it is shown that generally, only 50% of the downtime is directly attributed to the repair operation. The remaining time is allocated to reaching the machine, delays, preparation, diagnosis, performance, testing, and verification of repairs. Unfortunately, the substantial time spent reaching the machines, on-the-job delays, lack of coordination between units, breaks, etc., have imposed significant damages on the fleet's maintainability. These challenges, however, can be effectively addressed by allocating suitable light vehicles, refurbishing the existing fleet, and leveraging up-to-date technologies in the jurisdiction of information exchange and operational communication between direct personnel and repair support. Therefore, it is not considered a fundamental challenge.

Mechanical failures often are caused by wear and tear, structural defects, or improper assembly, whereas electrical failures typically stem from issues such as short circuits, component malfunctions, or power supply irregularities. Understanding these differences allows maintenance personnel to identify the root causes of failures and implement suitable solutions.

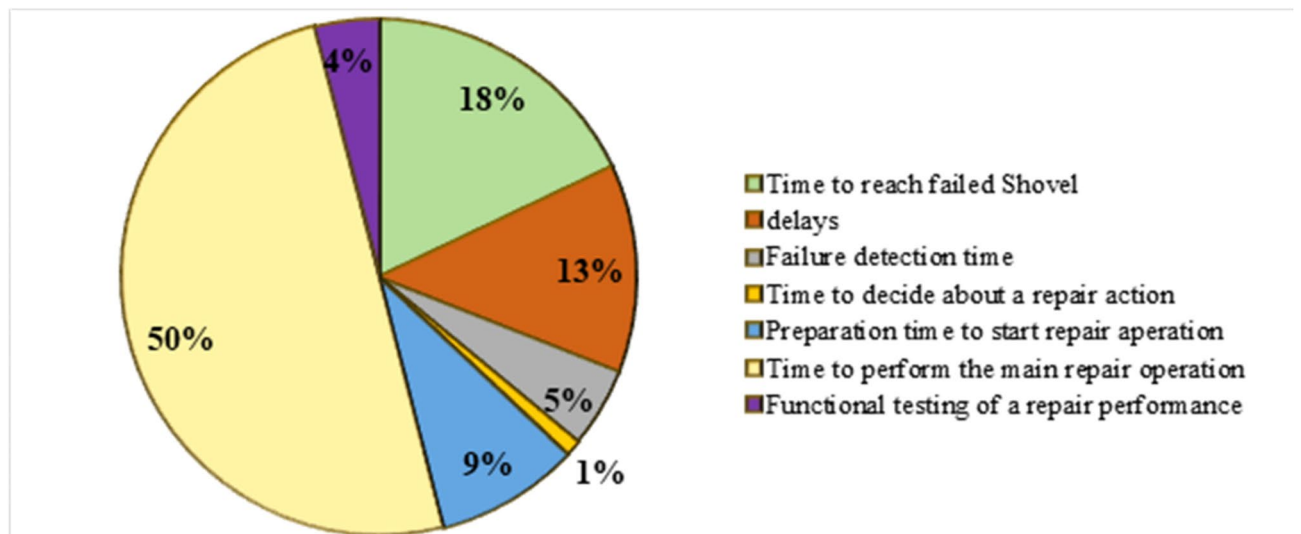


Fig. 9. The breakdown of maintenance operation components during the Sarcheshmeh copper mine shovel failures.

The performance of maintenance personnel in addressing mechanical and electrical failures differs due to the unique origins and characteristics of these failure types. In the case of mechanical failures, maintenance workers must focus on identifying and rectifying issues related to the physics of failure and the holistic design of machine and unit interactions. This often requires a strong understanding of mechanical systems, components, and physical inspection techniques to pinpoint the root cause and implement appropriate repair actions. On the other hand, addressing electrical failures demands a different set of skills and knowledge, as these issues stem from hidden and silent failure events. Maintenance personnel must demonstrate expertise in electrical systems, circuit diagnostics, and troubleshooting methods to accurately identify and resolve these problems. In essence, the difference in performance lies in the specific expertise and problem-solving abilities required to address the distinct nature of mechanical and electrical failures. Successfully tackling these challenges relies on the adaptability and diverse skill sets of the maintenance crew.

In mine maintenance shops, identifying the causes of electrical failures is primarily achieved through the use of computers and various monitoring systems which are mainly on-board or monitored from a control room. The time required to diagnose and repair these failures largely depends on the performance of these monitoring systems. However, when it comes to mechanical failures, the expertise and skills of the mine's maintenance personnel play a crucial role in identifying and addressing defects. This highlights the importance of having a versatile and highly skilled maintenance crew in mining operations. While advanced technologies such as eMaintenance platforms are essential for dealing with electrical failures, the human factor remains vital in handling mechanical issues.

Conclusion

In this paper, a wide range of historical repair data collected from a fleet of 10 rope shovels in the Sarcheshmeh copper mine was applied to perform a maintainability analysis. The overall investigation revealed that most of the maintenance operations are conducted in less than 45 h and 80% of them are finalized in less than a working shift. Nevertheless, due capital intensiveness of the shovel fleet and the high level of production loss in failed times, any attempts to reduce the present downtime duration and increase the fleet maintainability could boost the overall operation condition. For performing a deep investigation, a meticulous minute-by-minute examination of 50 types of shovel failures was carried out in the mine. The statistical analysis revealed that the time to reach the failed machine and associated delays are the two main reasons to increase the downtime duration. Both of them are directly related to the support and logistics of the mine workshop. Lower values of “failure detection time” and “Time to decide about a repair action” show that the maintenance crew of the mine is deeply skilled and could manage the maintenance operation properly. Finally, it was seen that only 50% of the downtime is spent on direct repair action and the rest of the time belongs to five different maintenance actions and delays. Inadequate provision of appropriate transportation (light and heavy-duty vehicles) for maintenance personnel resulted in substantial delays in reaching the shovels, consequently extending repair times.

Based on the identified bottlenecks, the following recommendations aim to streamline repair processes and reduce overall repair times:

- **Mobile Welding Unit:** Equipping shovel mechanics with a portable welding device has the potential to significantly reduce welding-related delays by 15–60 min per repair.

Enhanced Resource Allocation: Increasing the availability of light vehicles for the repair center personnel will improve their mobility and response times, expediting repair initiation.

- Time Management Training: Implementing comprehensive training programs focused on essential time management principles specific to repair procedures can potentially lead to a substantial decrease of at least 30% in overall repair times.
- Electrical Fault Diagnosis Complexity: Electrical faults, due to the absence of audible indicators, pose a greater diagnostic challenge compared to mechanical faults, res.

Data availability

The datasets used and analyzed during the current study available from the corresponding author on request.

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Author contributions

Mohammad Rezaei Dashtaki: Data curation, Writing – original draft. Ali Jandaghi Jafari: Investigation, Writing – editing. Seyed Hadi Hoseinie: Conceptualization, Methodology.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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